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ABSTRACT

Many foods have a yield stress that allows them to retain a desired shape at rest, but transition into a viscous fluid when being served or consumed. The determination of the yield stress of the food dictates how the foods are formed and packaged, how they are served, and how they are perceived when being eaten. Oscillatory shearing provides an ideal test protocol to map the rheology across a range of time and flow strength scales. We couple oscillatory shearing and an iterative recovery procedure to show that the yielding process is a continuous transition for two common yield stress foods. We show that unrecoverable processes from oscillatory tests are equivalent to the steady shear flow behavior. We show that this yielding behavior can be well approximated by a recently published model that treats yield stress materials as continuous viscoelastic fluids with a rate-dependent relaxation time and has parameters that can be obtained from the linear viscoelastic oscillatory frequency sweep and the steady shear flow curve.

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I. INTRODUCTION

One of the goals of food science is to design a user experience without the need for costly sensory analysis.[1] Understanding how the material responds during the eating, processing, packaging, and preparation processes involves complex combinations of tribology, rheology, and psychology as the food undergoes wear, friction, lubrication, bulk deformation and flow, and is perceived via optical, olfactory, and occasionally auditory senses.[2,3]

To understand the rheology of foods, namely, how they respond to stresses and strains, the material behavior must be probed across a wide range of timescales and deformation strengths. Small amplitude oscillatory shear (SAOS) tests probe timescales that relate to the processing and end-use of foods, but apply only weak perturbations and, therefore, miss the required flow strengths. Various flow strengths can be probed through the application of steady shearing, though determination of the steady-shear flow curve contains no relation to the timescales relevant to processing and end-use. To probe a wide range of timescales, however, as well as deformation strengths associated with manufacturing, processing packaging, biting, chewing, and swallowing, large-amplitude oscillatory shearing (LAOS) is utilized.[4–7] In LAOS, the amplitude and angular frequency allow for tuning of the flow strength and time scales, respectively. A deep understanding of the LAOS rheology would, therefore, provide guidance for the design of the parts of the user experience associated with the rheology, as well as allow for the creation of structure–property–processing relations.

Much study has been devoted to understanding responses to LAOS. When applying a large amplitude sine strain to a material, the stress response is nonlinear.[8] An early work by Harris and Bogie investigated the raw stress response of an applied strain wave.[9] They applied Fourier transforms (FT) to the stress response and presented the harmonic contributions. This Fourier analysis method of LAOS data treats the material response as a linear summation of potentially infinite basis states, and has become known as Fourier transform (FT) rheology, which has since been studied and developed extensively by Wilhelm.[10] It has been noted, however, that the Fourier analysis of stress responses lacks clear physical interpretation, which makes using these results difficult for design.[11]

Another early work by Philippoff introduced the idea of parametrically plotting the stress response vs the input strain in a Lissajous curve.[12] This analysis approaches the material response geometrically, by examining how the stress response relies on the input strain. In 2005, Cho *et al.* suggested a stress decomposition (SD) into so-called elastic and viscous stresses via symmetry arguments.[13] From this, definitions of even and odd functions were determined via the elastic Lissajous figures, where stress is plotted against the input strain, and

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viscous Lissajous figure, where stress is plotted against the strain rate. These even and odd functions described single-valued elastic and viscous stresses.

In 2008, Ewoldt *et al.* suggested the use of Chebyshev polynomials of the first kind as a way of describing the stress decomposition parameters, while also connecting the Fourier coefficients.[11] The Chebyshev description of the SD parameters, like FT rheology, describes a material response as a linear summation of potentially infinite basis states, but with Chebyshev polynomials as the basis, rather than harmonic trigonometric function.

LAOS has been used as an experimental probe of the nonlinear rheology of food for more than two decades.[14–21] Cheeses, yogurts, gels, doughs, chocolates, and pastes have all been studied with LAOS.[1,22,23] The SD and Chebyshev approaches are the most common ways of analyzing the response to LAOS of foods. Their inclusion in commercial rheometry software makes their use appealing and they continue to remain popular methods of analysis in food rheology.[1]

In 2011, Rogers and Lettinga showed that the symmetry assumptions on which the SD and Chebyshev approaches were based lead to unphysical interpretations for simple yield stress materials. A series of papers then introduced a new method of understanding LAOS that has been called the sequence of physical processes (SPP).[24]

The SPP was initially proposed as a qualitative method of interpreting LAOS data by decomposing it into a *sequence* of material responses, in direct contrast to the *linear summations* of FT and the Chebyshev approaches.[24] It was shown that a yield stress material response should be decomposed *sequentially* into an elastic response which gives way via yielding to viscous flow. This interpretation was supported by examining different regions of a Lissajous curve, showing that the slope of the stress with respect to the strain close to zero stress correlates with the linear regime storage modulus. Linear viscoelastic material parameters were, therefore, shown to appear in the nonlinear material responses. Following the interval of linear-like response, it was shown that the stress response essentially followed the same behavior as the steady-state flow curve, showing that LAOS can be viewed as a mapping between linear-regime frequency sweeps and steady-shear flow curves. Further, the ability to identify linear viscoelastic parameters implies that the material is close to its local equilibrium, which, when coupled with the observation that such features are observed at significantly large strains, further implies that strain has been acquired unrecoverably, and a new equilibrium position has been established.

Following the qualitative SPP, a quantitative version of the SPP was introduced in 2012[25] and developed in 2017.[26] The fully quantitative SPP approach, which has recently been made available as a standalone software package,[27] acknowledges a difference between the total and recoverable strain and is the only LAOS analysis that provides a method for estimating the recoverable strain from knowledge of the stress response to the total strain. However, the recoverable strain can only be determined within the SPP approach under conditions where the instantaneous elastic response is much larger than the instantaneous viscous response. Given the limited nature of the SPP's capability of mathematically determining the recoverable strain, Lee *et al.* developed an iterative experimental procedure to determine recoverable and unrecoverable strains during LAOS.[28]

The ideas of recoverable and unrecoverable strains go back at least as far as the studies of Weissenberg and Rideal[29] and Reiner,[30] and some even earlier tests on doughs measured and reported the recoverable strain after deformation.[31] However, experimental studies that measure the recoverable and unrecoverable strain have been sparsely reported. The iterative procedure developed by Lee *et al.* to determine the evolution of the recoverable strain[28] allowed for the definition of new metrics to describe the moduli in terms of recoverable and unrecoverable strain. Donley *et al.*[32] followed proofs presented by Tschoegl[33] and showed how the definitions of the storage and loss moduli in terms of the average storage of energy and rate of dissipation of energy, respectively, lead to an understanding of the commonly observed overshoot in G'' that has been typically associated with yielding.[34]

Many foods have a small but finite yield stress, allowing them to be spread onto other foods easily and providing the possibility of artistic sculpting for presentation purposes.[2] The presence of a yield stress is often engineered via formulation selection, so that the consumer experience is enhanced. While yield stress rheology has been studied extensively, there is still no widely accepted description of what yielding is, nor when it occurs.[35] For foods that require a yield stress for their function, there is little room to design the user experience, as the research and development process relies on trial and error rather than an approached based on the desired qualities.

A common conceptual description of yield stress fluids invokes the Oldroyd–Prager[36] formalism, which consists of considering the material behavior separately below and above the yield stress. Oldroyd was the first to describe the behavior below the yield stress as an elastic solid and as a Bingham fluid[37] above the yield stress. Yielding within the Oldroyd–Prager formalism is a discontinuous and instantaneous transition from one behavior to the other, though it should be noted that Oldroyd only ever considered the case where the yield stress was approached in a quasi-static manner.[36,38] In 2007, Saramito presented a model that addressed an outstanding continuity issue in Oldroyd's model, but retained the same formalism and physics.[39] While an abrupt yielding transition has often been invoked in the literature, experimental support for the concept over finite timescales remains lacking.

From a decomposition of strains into recoverable and unrecoverable components of simple yield stress fluids, Kamani *et al.* presented a simple continuous model of yield stress fluids that avoids the Oldroyd–Prager formalism.[40] Kamani *et al.*'s model describes yield stress materials as a continuous viscoelastic fluid with a rate-dependent relaxation time. This recently proposed model only requires parameters determined in a linear regime frequency sweep and a steady shear flow curve, and yet accurately predicts the LAOS behavior. It, therefore, supports the conclusions from the phenomenological SPP studies that LAOS can be understood as a mapping between these two rheological extremes. Because Kamani *et al.*'s model has been shown to accurately predict the rheology of hydrogels that are simple yield stress fluids, it is desirable to test it against multi-ingredient complex food systems.

In this work, we present the results of a study into the yielding of two common yield stress foods, Kraft-Heinz Cool Whip and Philadelphia cream cheese, via LAOS with an iterative recovery protocol. The manuscript is organized as follows: Sec. II introduces the two material systems and the experimental techniques; Sec. III presents the results of the experiments in terms of traditional and recovery rheology measures, and shows the connection between these measures and

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the steady shear flow curve. Predictions of Kamani *et al.*'s model are also compared to the experimental results in Sec. III. In Sec. III C, the implications of our study are discussed in terms of the SD and Chebyshev analysis methods with a particular focus on the symmetry assumptions.

II. METHODS AND MATERIALS

Cool Whip is a stabilized foam produced from hydrogenated vegetable oil, with added cream, and other ingredients, some of which themselves are yield stress materials, such as xanthan gum and guar gum.[41] Cream cheese is a concentrated colloidal suspension[42] manufactured from pasteurized milk.[43] Cool Whip and cream cheese are common yield stress foods where the yield stress is a desired property that allows the foods to hold their shape after being smeared or applied to other foods. Both materials typically undergo large amounts of strain in their use, from food preparation to consumption.[43] They represent two classes of yield stress foods, stabilized foams and concentrated colloidal suspensions, that have different microstructures and important length scales, and yet share many similarities in their rheology. Their testing, therefore, presents a microstructural-agnostic test of yield stress rheology.

Kraft-Heinz Cool Whip and Philadelphia cream cheese were obtained from local supermarkets and stored at 5 °C until testing. Cool Whip was tested within 2 weeks of thawing while cream cheese was tested within 2 weeks of opening the top foil. Each experiment was conducted at 5 °C.

All experiments were performed on an Anton Paar MCR 302 Rheometer under strain-controlled mode unless noted otherwise. A 25-mm plate geometry was used, with 220 grit sandpaper attached to both surfaces to avoid wall slip. The linear regime frequency sweeps, nonlinear flow curve, and traditional amplitude sweeps were all performed using standard functions inside the RheoCompass software.

Recoverable strain tests were performed by using the list function inside the steady strain measurement option in RheoCompass, in which each data point of oscillation is imported to the software with a resolution of 512 points per period, and a frequency of 1 rad/s. Each recovery test consists of four full periods of oscillation to allow for the steady alternating state to be reached, followed by a partial period of oscillation then immediately by a zero-stress step lasting 40 s. Subtracting the strain value at the end of the recovery interval from the total strain right before the application of zero-stress gives the recoverable strain. A graphical description of this process is shown in Fig. 1. The partial period of oscillation changes where the zero-stress step begins, depending on where in the period is being examined. The recovery points were located approximately every tenth of a second, related to the period length divided by the resolution multiplied by the points per step (time per period/points per period ∗ points per step). All tests were performed in the forward and reverse directions and averaged, to account for any instrument drift. Repeating this for half a period and mirroring the data, we construct a full period of recoverable and unrecoverable strains.

III. RESULTS AND DISCUSSION

A. Traditional rheological characterization

A traditional rheological characterization typically involves determination of the small-amplitude oscillatory response to varying frequencies and a measurement of the steady-shear flow curve. The

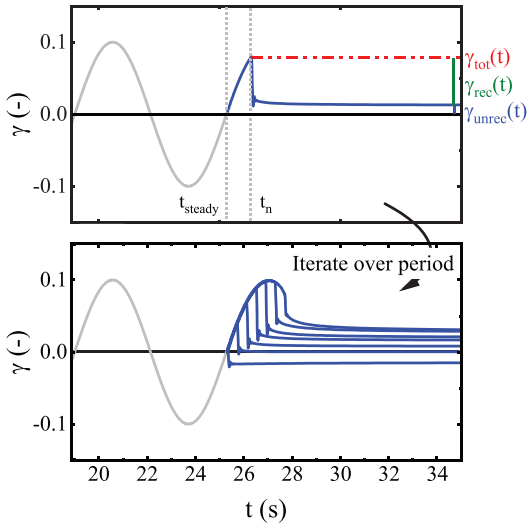


FIG. 1. Graphical description of a single recovery rheology test, as well as the iterative result of many recovery tests over the period.

results for the Cool Whip are shown in Figs. 2(a) and 2(b), respectively, and indicate that close to equilibrium the material behaves as a viscoelastic solid with a modulus between 3000 and 5000 Pa, while its steady shearing response is well described by the Herschel–Bulkley model with a dynamic yield stress close to 50 Pa. Similar observations are made in the traditional rheology of the cream cheese, which is shown in the Appendix and in Figs. 9 and 10.

Many popular methods of determining a yield stress or strain are obtained from an oscillatory amplitude sweep,[35] where the frequency is fixed, and progressively larger amplitudes are applied. The results of performing an amplitude sweep at 1 rad/s on the Cool Whip are shown in Fig. 3, with a range of measures that have variously been referred to as "the yield stress" in the literature. Similar observations for the cream cheese are shown in the Appendix. Six different ways of determining a yield stress are shown. The range of measures used in

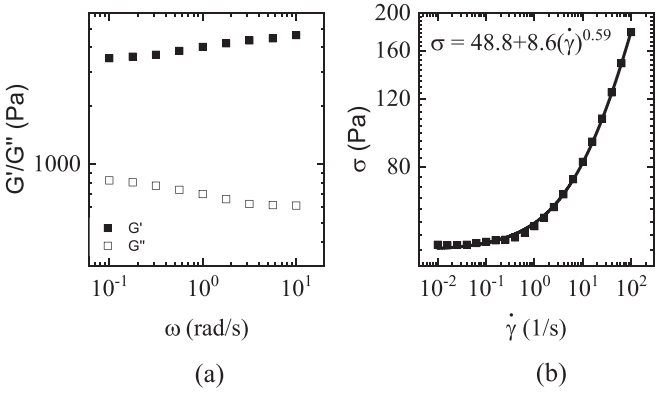


FIG. 2. (a) Linear regime frequency sweep of Cool Whip. Closed squares are the storage modulus, and open squares are the loss modulus and (b) steady shear flow curve of Cool Whip with Herschel–Bulkley fit, showing a yield stress of ∼50 Pa.

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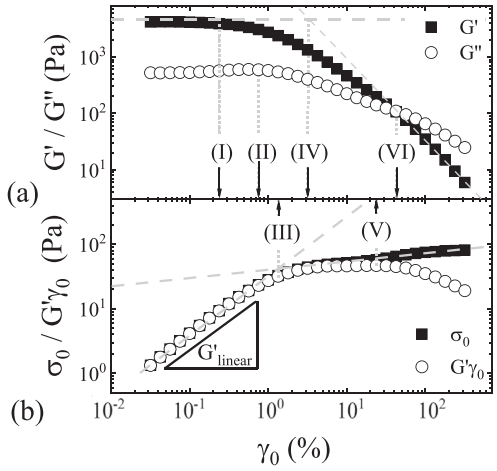


FIG. 3. (a) Amplitude sweep for Cool Whip at a frequency of 1 rad/s. Each arrow represents where "the" yield stress of the material could be determined, along with Table I for each value description. (b) Stress amplitude and $G'\gamma_0$ vs strain amplitude of the same amplitude sweep data.

the literature makes the question of determining yield stress difficult to answer unambiguously.[35]

For both materials at small amplitudes, $G' > G''$, seen in Figs. 3(a) and 10(a) as closed squares and open circles. As the amplitude of the applied strain increases, G' begins to decrease, and this deviation from linearity has been reported as one measure of the yield stress.[44] The intercept of power law fittings of G' at small and large amplitudes is another measure that has been reported.[45] Yet another point is when G'' reaches a maximum at intermediate amplitudes.[46] At larger amplitudes, the moduli cross, so that $G' < G''$. This crossing point has also been called a yielding point[47] or the flow point.[48]

The definitions and values of these various points are shown in Table I, along with two additional measures that require examination of the strain amplitude dependence of the stress amplitude[49] and the product of the storage modulus and the strain amplitude, shown in Figs. 3(b) and 10(b) as closed squares and open circles. This last term, the product of the storage modulus and the strain amplitude, has erroneously been referred to as the elastic stress.[50] It more accurately reflects a scaled measure of the average amount of energy stored, as will be discussed further below. The wide range of values commonly used in the literature leads to a clear and significant ambiguity in determining when yielding takes place and the value of the yield stress, as these measures span nearly a decade in stress and two decades in strain.

B. New rheological descriptions

In the amplitude sweeps, each datum reflects an averaged response over the entire period of oscillation, and reduces the often complex waveforms to two values. Lissajous curves are a way of looking at each of these amplitudes by parametrically plotting the stress response vs the applied strain or rate. In Fig. 4, the stress response is plotted against the applied strain for each material in the range of $\gamma_0 = 10\%$ to 100%. In addition to the raw waveforms, regions of similar elastic modulus are highlighted, and specifically show solid lines with slopes equal to that of the linear-regime elastic modulus. This correspondence is evidence of a nearly linear viscoelastic response *within* the nonlinear response close to zero stress. In addition, qualitative matching with the sequential understanding presented by the SPP is seen, in which the linear regime response can be seen in the Lissajous data. However, this equilibrium response now occurs about a non-zero strain, implying that significant unrecoverable strain has been acquired.

While a qualitative SPP analysis of the Lissajous figures provides some insight into the physics occurring during a small interval of time, it does not account for the full amplitude dependence of the dynamic moduli. Building on proofs presented by Tschoegl,[52] the work of Donley *et al.* showed that the dynamic moduli at all amplitudes can be understood in terms of average amounts of stored energy and the rate of dissipated energy normalized by the total strain amplitude,[32] as shown in the following equations:

$$G'(\omega) = \frac{4[W_{\text{stored}}(\omega)]_{\text{avg}}}{\gamma_0^2} = \frac{2[\gamma(t)\sigma(t)]_{\text{avg}}}{\gamma_0^2}, \tag{1}$$

$$G''(\omega) = \frac{2[\dot{W}_{\text{diss}}(\omega)]_{\text{avg}}}{\gamma_0^2} = \frac{2[\dot{\gamma}(t)\sigma(t)]_{\text{avg}}}{\omega\gamma_0^2}. \tag{2}$$

Equation (1) makes clear why calling the product of the storage modulus and the strain amplitude "the elastic stress" is problematic. This product, if it is to be used, is better interpreted as a scaled ratio of the stored energy to the strain amplitude.

TABLE I. Yield stress and strain values determined via oscillatory amplitude sweep and Herschell–Bulkley flow curve fitting.

			Cool Whip		Cream cheese	
Label	Fitting	Reference	σ_y (Pa)	γ_y (%)	σ_y (Pa)	γ_y (%)
I	Deviation from linearity in G'	Mezger[44]	9.2	0.25	71	0.134
II	G'' overshoot [max(G'')]	Donley *et al.*[46]	23	0.75	147	0.318
III	Intersection of power-law equations, σ_0 vs γ_0	Mason *et al.*[49]	30	1.2	410	1
IV	Intersection of power-law equations, G' vs γ_0	Rouyer *et al.*[45]	44	3.1	800	6.2
V	Elastic stress [max($G'\gamma_0$)]	Walls *et al.*[50]	57	24	791	10
VI	G'–G'' crossover	Shih *et al.*[47]	64	42	749	42.4
⋯	Herschel–Bulkley fit	Herschel and Bulkley[51]	48	7.5	731	40

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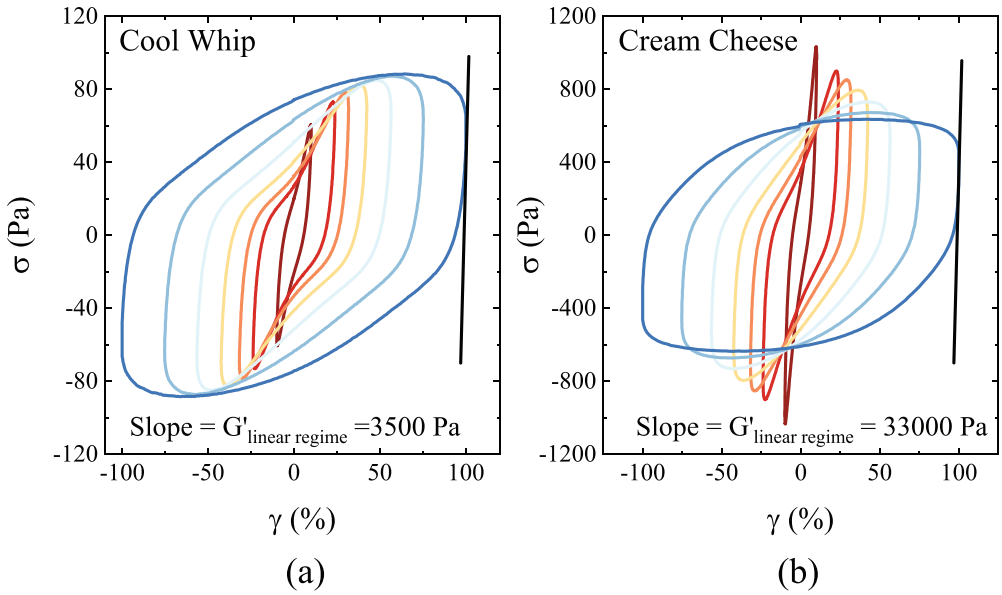


FIG. 4. Lissajous curves of waveforms for (a) Cool Whip data and (b) cream cheese in the nonlinear regime on amplitude sweep seen in ranging from $\gamma_0 = 10\%$ to $\gamma_0 = 100\%$. The slope of the lines represents the linear regime storage modulus.

The work of Donley *et al.* further showed that the dynamic storage modulus can be completely determined in terms of the recoverable strain, and was explicit in naming this G'_{solid}, where the subscript "solid" relates to recoverable strains, as shown in in the following equation:

$$G'_{\text{solid}}(\omega) = \frac{4\left[W_{\text{stored,solid}}(\omega)\right]_{\text{avg}}}{\gamma_0^2} = \frac{2[\gamma_{\text{rec}}(t)\sigma(t)]_{\text{avg}}}{\gamma_0^2}. \quad (3)$$

Because an arbitrary transient nonlinear response can be decomposed experimentally into recoverable and unrecoverable strains, there are also two strain rates that provide distinct energy dissipation mechanisms. The dynamic loss modulus, which is proportional to the total amount of energy dissipated, can therefore be recast as the sum of two terms, one of which is defined in terms of the recoverable strain rate, while the other is defined by the unrecoverable strain rate, G''_{solid} and G''_{fluid}, as shown in in the following equations:

$$G''_{\text{solid}}(\omega) = \frac{2\left[\dot{W}_{\text{stored,solid}}(\omega)\right]_{\text{avg}}}{\omega\gamma_0^2} = \frac{2[\dot{\gamma}_{\text{rec}}(t)\sigma(t)]_{\text{avg}}}{\omega\gamma_0^2}, \quad (4)$$

$$G''_{\text{fluid}}(\omega) = \frac{2\left[\dot{W}_{\text{stored,fluid}}(\omega)\right]_{\text{avg}}}{\omega\gamma_0^2} = \frac{2[\dot{\gamma}_{\text{unrec}}(t)\sigma(t)]_{\text{avg}}}{\omega\gamma_0^2}. \quad (5)$$

These component moduli are shown for both Cool Whip and cream cheese in Fig. 5. At small amplitudes $G' > G''$, which is common in yield stress fluids. Additionally, the solid-like dissipation of energy determined by the recoverable strain rate, G''_{solid}, dominates the fluid-like dissipation of energy determined by unrecoverable strain, G''_{fluid}. This is a clear indication that both Cool Whip and cream cheese act like a viscoelastic solid at small amplitudes, as almost all strain is acquired recoverably. At a point around $2-5\%$, the energy dissipation due to acquisition of unrecoverable strain becomes larger than that associated with recoverable strain, and leads to G''_{fluid} becoming larger than G''_{solid}. Then at a larger amplitude around 30%, G''_{fluid} also overtakes G'_{solid} becoming the dominant energetic contribution.

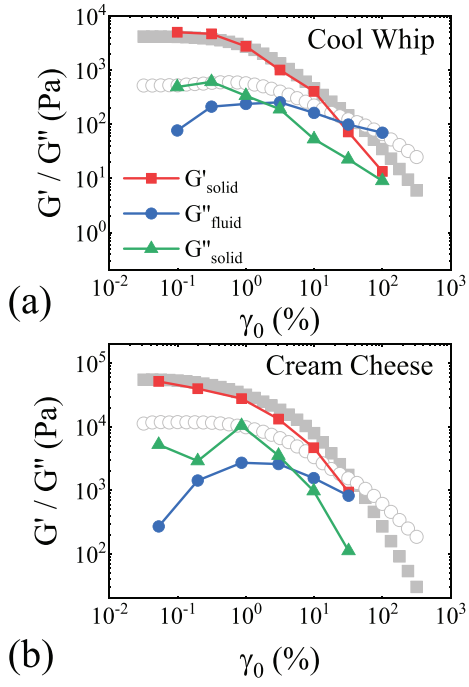


FIG. 5. Traditional amplitude sweep, overlaid with component moduli determined from recovery rheology experiments for (a) Cool Whip and (b) cream cheese.

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The acquisition of unrecoverable strain may be taken as a necessary condition for the determination of when yielding occurs. The non-zero value and smooth and continuous increase in G''_{fluid} at small and intermediate amplitudes indicate that yielding is a smooth and continuous process.

The unrecoverable metrics obtained from oscillatory testing can be compared directly to the steady shear flow curve. To do so, the average dissipation of energy associated with unrecoverable strain is normalized by the unrecoverable amplitude, rather than the total strain amplitude, to create a new energetic metric that is defined entirely on the unrecoverable flow behavior of the material, named here as G''_{unrec},

$$G''_{\text{unrec}}(\omega) = \frac{2(\dot{\gamma}_{\text{unrec}}(t)\sigma(t))_{\text{avg}}}{\omega(\gamma_{0,\text{unrec}})^2}. \tag{6}$$

Shown in Fig. 6 is the steady shear viscosity, determined by dividing the steady shear stress of Fig. 2(b) by the shear rate, the Herschel–Bulkley description of the steady shear data, and the energy dissipation term normalized by the unrecoverable strain from recovery tests, G''_{unrec}. The agreement between the three measures of viscosity is excellent, indicating that the unrecoverable strain measure determined in LAOS provides the same information as the steady-shear flow curve. The equality of the different measures also signifies interesting physics for yield stress foods. The energy dissipation term normalized by the unrecoverable strain from recovery tests, G''_{unrec}, is shown to be equal to the steady-shear viscosity when plotted against the total applied strain rate. In the oscillatory tests, the total strain rate includes a contribution from the recoverable strain rate, $\dot{\gamma} = \dot{\gamma}_{rec} + \dot{\gamma}_{unrec}$. The equivalence displayed in Fig. 6, therefore, implies that the viscosity associated with acquisition of unrecoverable strain is decreased by rapid acquisition of recoverable strain. Elasticity enhances the yielding.

The simple viscoelastic model developed by Kamani *et al.* includes a strain-rate dependent relaxation time.[40] The model combines a recoverable component that is a Kelvin–Voigt viscoelastic solid with parameters defined by the linear-regime frequency sweep shown in Fig. 2(a) and a steady shear flow curve that has a Hershel–Bulkley viscosity, as suggested by the data of Fig. 2(b). The key difference between the model of Kamani *et al.* and the retarded Maxwell model,[53] or those of Jeffreys or Oldroyd,[36] is the linking of the recoverable and unrecoverable components. In the Kamani *et al.* model, the viscosity of the unrecoverable component is dependent on the total strain rate rather than just the unrecoverable strain rate. Kamani *et al.* presented a full tensorial version of the model in addition to the simpler 1D version that provides a connection between the shear components of the stress and strain tensors. We restrict our study here to the simpler 1D version of the model, which is shown in Eqs. (7) and (8),

$$\sigma + \lambda(\dot{\gamma})\dot{\sigma} = \left(\frac{\sigma_y}{|\dot{\gamma}|} + k|\dot{\gamma}|^{n-1}\right)\left(\dot{\gamma} + \frac{\eta_s}{G}\ddot{\gamma}\right). \tag{7}$$

The rate dependent relaxation time, $\lambda(\dot{\gamma})$, is

$$\lambda(\dot{\gamma}) = \frac{1}{G}\left(\frac{\sigma_y}{|\dot{\gamma}|} + k|\dot{\gamma}|^{n-1} + \eta_s\right). \tag{8}$$

This model is easy to use, as it only requires parameters determined from the tests shown in Figs. 2(a) and 2(b). The elastic modulus, G, and the solid viscosity, η_s, can be found via the linear regime frequency sweep as $G = G'(\omega)$ and $\eta_s = G''(\omega)/\omega$ both at a frequency of 1 rad/s. The unrecoverable strain rate terms, σ_y, k, and n are all determined by a Herschel–Bulkley fitting of the flow curve. Here, σ_y is the Herschel–Bulkley yield stress, k is the consistency, and n is the shear thinning exponent.

The predictions of this model for the components of the dynamic moduli in the amplitude sweep are compared to the experimental results in Fig. 7, where the experimental data are shown as symbols and the model predictions are solid lines. The model does well

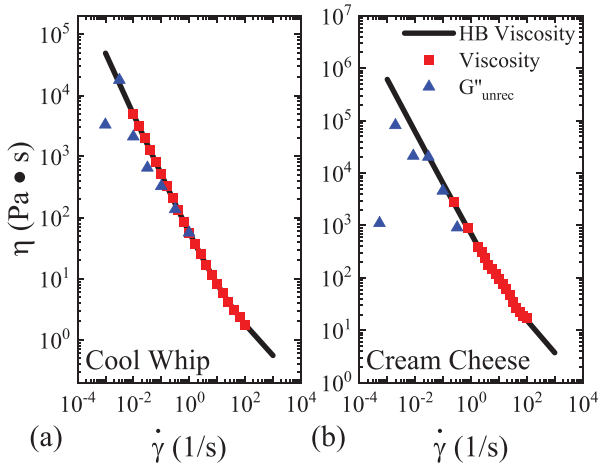


FIG. 6. Herschel–Bulkley viscosity compared to component fluid moduli normalized by the unrecoverable strain amplitude for (a) Cool Whip and (b) cream cheese.

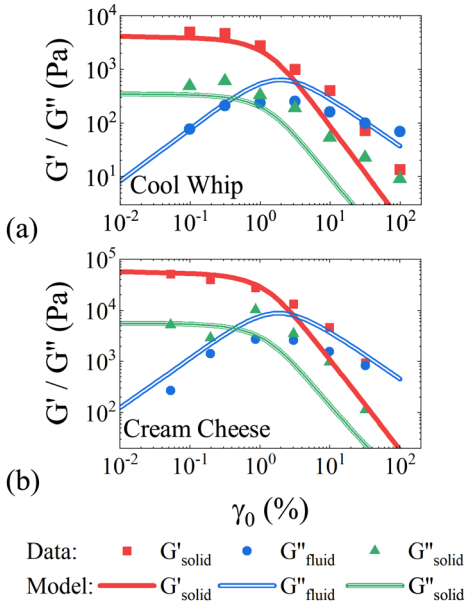


FIG. 7. Simple model predictions and experimental results as lines and symbols, respectively: G'_{solid} (red squares and top line starting on left), G''_{solid} (green triangles and middle line starting on left), G''_{fluid} (blue circles, and bottom line starting on left). Results for (a) Cool Whip and (b) cream cheese.

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describing the rheology of both the Cool Whip and the cream cheese, with many qualitative and quantitative similarities. In both the model and the experiments, $G' > G''$ at small strain amplitudes, while the moduli crossover at larger amplitudes. The model is also able to capture the behavior of the component moduli, G'_{solid}, G''_{solid}, and G''_{fluid} across all amplitudes. The model and the experiments show that at small amplitudes the energy dissipation is due to acquisition of recoverable strain, while at large amplitudes it is due to the acquisition of unrecoverable strain.

Although the Kamani *et al.* model does a good job of capturing most of the overall qualitative behaviors of the two yield stress foods, there are a few features of the model that are not exhibited by the data, and some quantitative differences. The experimental data do not show an overshoot in G'', which the model predicts. The model also predicts that the dynamic moduli should cross at a smaller strain amplitude than is observed experimentally. Further, the large amplitude slopes of the dynamic moduli are under-predicted by the model. Despite these differences, studies that measure the recoverable and unrecoverable strains can be used to inform the ways in which this model and others could be altered to better reflect the complex rheology of multicomponent foods, taking us closer to the design of their rheology.

The model assumes a Herschel–Bulkley steady shear response and a Kelvin–Voigt linear viscoelastic behavior. The frequency dependence of the linear viscoelasticity is therefore unlikely to be faithfully reproduced by this model, but the fact that it does so well with the large-amplitude oscillatory shearing is instructive. It shows that LAOS can be understood, even in systems as complex as foods, as being a path between these two limits. The form of the linear viscoelastic response could be enhanced adding additional modes to match the SAOS responses, but we have refrained from doing so here to focus more on the transient nonlinear physics.

C. Implications for popular LAOS analyses

LAOS has been used to probe the transient nonlinear response of numerous foods, and in most cases, the data have been analyzed via either Fourier transformation or the Chebyshev description of the SD parameters. The SD uses mathematical arguments to decompose the stress into so-called elastic and viscous stresses. Geometrically, these parameters are formed by looking at stress vs strain or stress vs strain rate in the elastic or viscous Lissajous figures. At the same total value of strain or rate, the stress values are averaged to determine the elastic or viscous stress. Arguments have been previously presented[25,54] regarding limitations of the SD assumption, highlighting the caution that needs to be applied when using these approaches. One case that has yet be discussed is the simple case of the Maxwell model of viscoelastic liquids. A mechanical analogue of the Maxwell model is made up of a Hookean spring and Newtonian dashpot in series, so that the stress in each component is the same, while the strains are additive. An indication of the limitations of the SD and the Chebyshev analyses is that these schemes create a distinction between the total stress and the elastic and viscous stresses for even the simplest viscoelastic liquid model that do not have a physical interpretation.

In addition to the physical arguments presented by Rogers and Lettinga[25] and Poulos *et al.*,[54] we present experimental evidence of why the interpretations of the SD and Chebyshev analyses should only ever be applied with extreme caution, if at all. One of the main contributions of this work is the (re)introduction of the use of recoverable and unrecoverable strains to understand transient nonlinear rheology of foods. We show how the full amplitude sweeps of two yield stress foods can be understood as a transition from energy dissipation by recoverable processes at small strains to energy dissipation by unrecoverable processes at large strain amplitudes. The work of Lee *et al.* showed that the structure of simple viscoelastic materials is related to the recoverable strain measured in the same way we have used in this study.[28] Earlier work of Ferry[53] presented similar arguments in a more general context for polymeric solutions. The key concept of these works is the connection between *recoverable* strain and a structural measure. Central to the SD and Chebyshev analyses is the assumption of a connection between the *total* strain and structural measures. Figure 8 shows how the recoverable strain changes with the total strain for both foods studied here. It is clear that the structurally relevant recoverable strain is not a single-valued function of the total strain. Averaging over stresses at the same strain, therefore, conflates information from distinct structural states evidenced by the different recoverable strain. In stark contrast, access to the fundamental physics at the heart of transient nonlinear rheology is provided by experimentally separating the recoverable and unrecoverable processes, as we do here.

The SD is a mathematical approach to estimating stress contributions that can be checked by measuring them experimentally. Stress jump tests have been used since the work of Mackay and Kaffashi in 1995[55] to experimentally determine elastic and viscous contributions in complex fluids. In a stress jump test, the shear rate is set to zero as quickly as the instrument allows, and the stress decay is measured. The elastic contribution is determined via extrapolation back to the time the rate was reduced to zero, where viscous stresses will have been reduced to zero. O'Brie and Mackay[56] used the same experimental technique to study shear thickening, while Dullaert and Mewis[57] employed stress jumps to study the stress contributions in thixotropic suspensions. More recently, Ong *et al.*[58] developed the technique further, and included iteratively performed flow reversals with stress

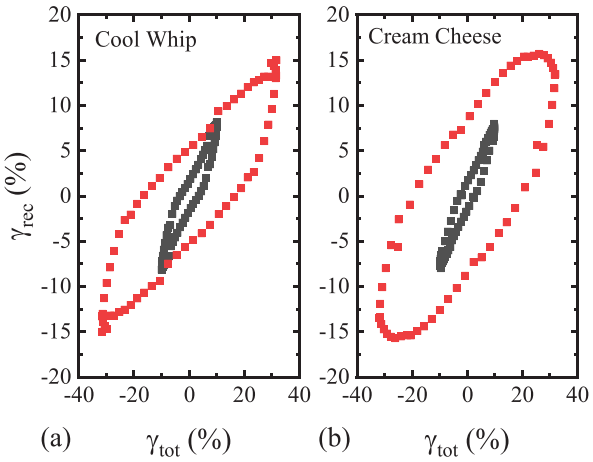


FIG. 8. Parametric plotting of recoverable strain vs total strain for (a) Cool Whip and (b) cream cheese. The double-valued nature of the recoverable strain at each value of total strain indicates that each material is in distinct rheological states at the same total strain. Averaging stress responses at the same total strain will, therefore, conflate different physics.

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jumps to determine contact, Brownian, and hydrodynamic stresses in their study of the LAOS response of dense colloidal suspensions.

When viewed in light of these prior experimental studies and the current experimental study, the symmetry assumptions at the heart of the SD and Chebyshev analyses can be said to have been shown not to hold in the general sense. Additionally, the iterative techniques employed here point the way toward a powerful experimental framework for understanding the transient nonlinear rheology of food and other soft materials. These iterative techniques have been used here to determine strain components, while other studies have used similar approaches to determine stress components. These iterative approaches are appealing for food rheology studies because they do not require complex mathematics nor access to, and the use of, a mathematical solver program, such as MATLAB.[1]

IV. CONCLUSIONS

The transient nonlinear rheology of two yield stress foods with different microstructures was studied by combining LAOS and an iterative recovery procedure. It was shown that Kraft-Heinz Cool Whip and Philadelphia cream cheese share similar linear viscoelastic and steady shearing responses, which indicate that they are both viscoelastic solids under small stresses, and have Hershel–Bulkley flow behaviors.

The amplitude sweeps of the two foods, which can be used to define six different measures of the yield stress, also show similar features, including amplitude-independent dynamic moduli at small amplitudes with $G' > G''$. At larger amplitudes, both moduli decrease and eventually cross at strain amplitudes around 30%. A visual inspection of the Lissajous figures collected during amplitude sweeps shows regions of near constant slope, indicating the presence of a linear elastic response *during* a period of LAOS.

Iterative recovery tests were used to separate the dynamic loss modulus into two components, one each representing normalized energy dissipation from recoverable and unrecoverable strains. It was shown that at small amplitudes, energy is preferentially dissipated by recoverable processes, while at larger strains energy is dissipated predominantly through unrecoverable processes. The energy dissipation term from unrecoverable processes can be rescaled by the unrecoverable strain amplitude, rather than the total strain amplitude. This new term, G''_{unrec}, was shown to be equivalent to the viscosity determined from steady shearing, showing a clear connection between steady shear and LAOS measures.

The amplitude sweep results, including the two components of the dynamic loss modulus that relate to recoverable and unrecoverable processes, were favorably compared to a recently published continuous model for yield stress fluids from Kamani *et al.*[40] The model predicts responses that are qualitatively similar to the experimental measures, allowing multiple measures of yield stress shown in Table I to be defined, but contains only one "true" yield stress, which is obtained from the Herschel–Bulkley fit to the flow curve. This addresses the difficulty expressed elsewhere[35] for defining the conditions that lead to yielding and suggests that "the" yield stress may be a material property that manifests differently across a range of situations.

Finally, the recovery measures were viewed in terms of the implications for common analysis methods for LAOS of foods. It was shown that the recoverable strain, which has previously been shown to correlate with the structural measures, is not a single value function of the total strain. The iterative techniques used here and elsewhere that determine stress and strain components point the way toward a powerful *experimental* paradigm for understanding the transient nonlinear rheology of foods and other soft materials. This comprehensive approach is appealing because it does not require complex mathematical analysis[1] and can be applied to any protocol. We have shown the ability to clearly link material responses across a range of tests, presenting a unified rheological picture. We, therefore, expect this approach to facilitate the formation of structure–property–processing relations and the rational design of user experiences by control of the nonlinear rheology.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

APPENDIX: COMPLIMENTARY DATA FOR CREAM CHEESE

The equivalents of Figs. 2 and 3 for Philadelphia cream cheese are shown in Fig. 9. Figure 9(a) is the linear regime frequency sweep

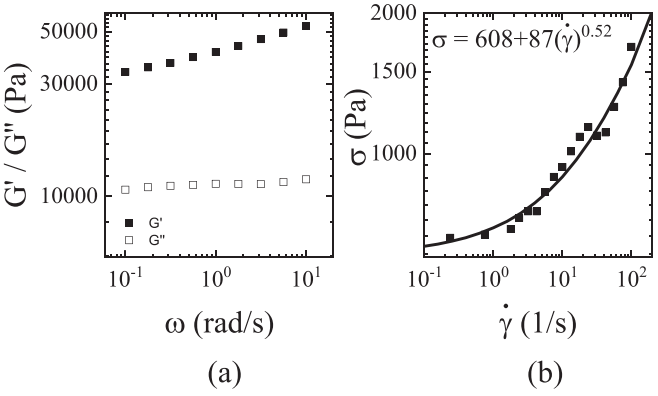


FIG. 9. (a) Linear regime frequency sweep of cream cheese. Closed squares are the storage modulus, and open squares are the loss modulus and (b) steady shear flow curve of cream cheese with Herschel–Bulkley fit, showing a yield stress of ∼600 Pa.

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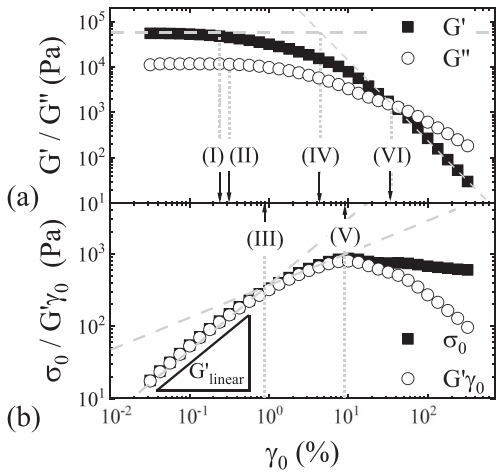


FIG. 10. (a) Amplitude sweep for cream cheese at a frequency of 1 rad/s. Each arrow represents where "the" yield stress of the material could be determined, along with Table I for each value description. (b) Stress amplitude and G'_{γ_0} vs strain amplitude of the same amplitude sweep data.

with the storage modulus larger than the loss modulus, implying the linear viscoelastic behavior is that of a viscoelastic solid. Shown in Fig. 9(b) is the steady shear flow curve determined as a series of constant rate tests after 1800 s. Shown in Fig. 10 are the same yield stress determinations as Fig. 3 and numerically seen in Table I.

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